Grant Proposal Activity

You are biochemistry experts employed by the **Susan G. Sneedleworth Foundation** (or SugSneeF). The foundation has an annual budget of $**7,500,000** to be spent on cancer-fighting projects of all kinds.

This year, you have received six grant proposals, and it is your responsibility to read and carefully assess their viability. In the end, you will be making funding recommendations that the chair of your foundation will use when deciding how to apportion this money.

Here's what to do:

1) Read through all six proposals
    - You may want to split this up so that each person is the "lead" for one or two proposals.
    - You have been hired because of your subject knowledge (Molecules & Life) and your general familiarity with scientific "habits of mind. Make sure to put these to use as you read! Judging these proposals might require a calculator, scratch paper, or some work.

2) Prepare a brief "report" for each proposal, explaining why it should be funded (or not).
    - This report doesn't have to be written down but should include 1 or 2 major points explaining why the project should or should not be funded.
    - Make sure you can cite specific reasons in your decision – you may be asked to defend these later.

Important points:

- You don't **have** to spend all the money. Make sure you're only funding projects you really thing deserve it.
- You can't give an applicant less money than they ask for. It's all or nothing!
- This is not a closed-book activity. Read through your notes about cancer, the central dogma, probabilities, the codon code, or anything else that might be relevant.

PANCREAX

New York City is home to more than 3,700,000 adults over the age of 40 - the cohort for which the risk of pancreatic cancer is highest. Because pancreatic cancer does not present with symptoms during the early stages, it is usually diagnosed only when it has progressed to stage 3 or 4 and is resistant to treatment.

Pancreax is pleased to present **PCX-I: a blood test for 16 biochemical markers that are elevated in the presence of pancreatic cancer.** As indicated by our recent clinical trial, PCX-1 is a highly sensitive test, accurately identifying 100% of participants with early-stage pancreatic cancer.

	Participants with stage I Pancreatic cancer (n = 48)	Control participants (n = 406)
Positive PCX-I Tests	48	5
Negative PCX-I Tests	0	401

Cancer patients were drawn from a pool of individuals at Mt. Sinai Tisch Cancer Center who had stage I cases of pancreatic cancer as confirmed by pancreatic biopsy. Control participants were recruited from the general population and were not currently undergoing treatment for any cancer. All control participants who tested positive were subsequently biopsied for pancreatic cancer and confirmed to be healthy.

Pancreax is committed to providing fair and equitable access to both emergency medical care and preventative medicine. *That's why we have partnered with NY Health to make preventative pancreatic cancer screening tests a part of every New Yorker's annual physical. With your funding assistance, we aim to make tests free-to-patient for all New York City residents 40 years or older who receive Medicare or Medicaid assistance.*

Given the extremely strong findings of Pancreax's in-house study, we feel that these tests are going to bring **peace of mind** to the many New Yorkers who test negative and offer a **road map to health** for those who test positive.

Budget request:

We are requesting ***$2,500,000*** to make free screenings available to all Medicare and Medicaid recipients – roughly 850,000 New Yorkers – per year.

Budget Breakdown	
Reagents and materials	$250,000
Blood sample delivery & pickup	$90,000
Diagnostic staff salaries	$1,550,000
Materials control & biohazard disposal	$180,000
Equipment & maintenance	$225,000
Office & laboratory space	$205,000

This donation will be matched dollar for dollar by NYC health, enabling us to provide coverage for less than $6 per test.

NOTE: Pancreax does not produce therapeutics for pancreatic cancer – only diagnostic tools.

For patients who test positive, physicians recommend scheduling a pancreatic biopsy to confirm the diagnosis. This will require referral to a gastroenterologist, as well as both preliminary and follow-up visits to discuss results. Pancreatic biopsies cost $3,500 on average.

From NY Health CEO Arden Gilai:



If caught early, pancreatic cancer is a surgically treatable illness. However, at the time of diagnosis most pancreatic cancers have metastasized. The five-year survival rate after a diagnosis of pancreatic cancer is only 12%. Nearly 400 New Yorkers over 40 develop pancreatic cancer every year. By subsidizing early screenings, New York City can do its part to identify and treat this serious illness.

While Pancreax and NY health remain committed to assuring access to affordable and effective healthcare, the funding requested in this proposal does not cover the cost of biopsy or diagnostic treatment for patients with positive PCX-1 test results. Because PCX-I is not currently recognized as a diagnostic tool by Medicare/Medicaid, these procedures may not be covered by insurance.



Dear friends,

We are a health advocacy and lobbying group which works to uncover the most dangerous airborne chemicals and remove them from the household items that emit them.

We are proud to report that we have conducted an extensive survey of New Yorkers, with over 65,000 responses gathered through a combination of street interviews, mail questionnaires, and phone calls. Respondents were asked about their experience with a list of **common household chemicals** as well as whether they had been diagnosed with **lung cancer**. This represents a new approach to identifying potentially carcinogenic substances, and we hope that it will provide valuable information for future regulations.

In this study, we found that survey-takers who were diagnosed with lung cancer were **significantly more likely** to report having purchased or used **Brühr P^3** (Premium Paint & Primer) in the past 5 years. *The full list of products asked about is attached on the following page.* Based on this, we commissioned a thorough investigation of the product by the independent chemical analysis company AnalyTec. *We've attached a copy of this report to our proposal, as well*. They found that one of the organic compounds present (the volatile organic compound **BtDCH3**) is known to interact with the **EGFR** gene and protein. EGFR, or epidermal growth factor receptor, is a protein responsible for signaling cell division. The EGFR gene is a known oncogene, and EGFR mutations are present in many lung cancers.

We think the responsible action would be to remove P^3 from the market, and since Brühr refuses to take our calls, we aim to pressure Albany to take steps to that effect.

We are asking for **$1,500,000** to launch a public health advocacy and lobbying campaign aimed at getting state lawmakers to ban the sale of Brühr paints until the carcinogenic **BtDCH3** compound is removed. We envision this campaign as covering two simultaneous strategies:

1. **Educate the public**
   Through online and TV commercials, we will highlight the link between **P^3** and **lung cancer**, with an emphasis on the risks associated with using this brand of paint, as well as how to pick paints that are low- or no-VOC options.
2. **Direct lobbying**
   We will send representatives to Albany to meet with state senators to discuss options for regulating the sale or the VOC content of **P^3** paint, as well as other paint brands which are found to include similar chemical compounds.

The full list of products surveyed – significant p-values indicate that respondents with lung cancer were more likely to have this item in their homes

Item	P-value
Mercury Scrubber™ kitchen cleaning powder	0.438
Lime-A-Way™ Shower Cleaner	0.771
Mr. Drypants™ Home Dry-Cleaning Kit	0.982
Flooral Bouquet™ Varnish	0.213
Brühr P³ (Premium Paint & Primer) ™	**0.042**
Noxox™ Drain Cleaner	0.515
Nailoff™ nail polish remover (with acetone)	0.661
Nailoff™ nail polish remover (without acetone)	0.723
Har-D-Boy™ Dishwashing Detergent	0.848
Tim the Grout Duck™ Grout	0.761
HoleBGone™ Spackle	0.376
Kaspar™ foam mattress	0.825
Keepit Roach Free™ spray roach killer	0.851
KopyKat™ printer toner	0.244
Wind-O™ ammonia-based glass cleaner	0.098
Hubert Humprey™ hard drive cleaner	0.811
Bloono™ foam insulation	0.325
Shino™ chrome insulation	0.448
DoubleBind™ epoxy glue	0.323
NardWarg™ rubber cement	0.761
Ocean Blast™ blue fireproof shower curtains	0.404
Heat Gust™ waterproof fire extinguisher	0.385
Whistle Clean™ laundry detergent	0.656
Wrex-Bex™ spray-on hat cleaner	0.628

To: Forward Health Advocates

Re: chemical analysis of supplied sample

Dear Engbert,

As per your previous email, here is a very abbreviated summary of our findings:

Detailed chemical analysis of the Brühr P³ (Premium Paint & Primer) interior paint (color type: elephant's breath) identified several VOCs (volatile organic compounds). These compounds vaporize easily to become airborne, and some present health risks when breathed in higher concentrations. The presence of VOCs is not necessarily evidence of a specific risk. Most house paints contain them, as do many common cleaning liquids, rugs and upholstery, and adhesives. They can also be produced through cooking and by natural sources such as plant matter.

We did uncover one VOC which may be of interest: **BtDCH3**, which has been studied in conjunction with the transmembrane protein EGFR, which regulates cellular replication. Although it is too large to diffuse into cells, **BtDCH3** binds to the extracellular regions of EGFR directly, mimicking a cellular growth signal and activating the protein's signaling cascade. In a relatively recent study, suspended cellular samples exposed to this chemical *in vitro* undergo rapid cellular division.

I hope this will suffice. Our complete 34-page report has been mailed to your office.

Best,
-Stroganoff T. Wizeborg (Chief Scientific Officer, AnalyTec)

phylogen

October 18, 2024
Naomi Belgrano - Vice president

To whom it may concern,

I am excited to share our preliminary findings on the efficacy of Floxoquil (also known as reminovabir), an mRNA vaccine for the Epstein-Barr virus.

Phylogen is highly respected in the pharmaceutical sector. Our research has led to the development of many effective therapeutic medications including Refloxavil, Supraminotab (marketed in Canada as Blistoflash), Hydroquildone, Hydroquildone-XL, and Hydroquildone-PM (marketed in Canada as Sleepofast). We have a proven track record of bringing therapeutic compounds to market and have recently expanded into vaccine production using our proprietary mRNA platform.

Epstein-Barr virus (also known as mononucleosis, mono, or the kissing disease) is a common respiratory virus. Like other herpes viruses (yes, mono is herpes), Epstein-Barr virus (EBV) infections are lifelong, though they are only symptomatic for a few weeks or months during the early stages of infection. Significant ongoing research links EBV infections to increased risk of nasopharyngeal cancers, lymphomas (including Hodgkin's lymphoma), and stomach cancer.

In the USA, most people are infected with EBV by the end of their teenage years. We estimate that for every 100,000 EBV infections prevented, as many as 8,800 cases of later-life cancer may be avoided.

Early-phase trials with mouse models demonstrate a high level of protection from a single dose of Floxoqul (i.e., reminovabir – see **Figure 1** on the next page).

We are seeking a grant of $3,000,000 to conduct phase 1 clinical trials. Phase 1 trials are aimed at testing safe dosages, cataloging side effects, and calibrating vaccine delivery methods in human subjects. They do not test vaccine efficacy and are conducted on healthy individuals. Phase 1 trials are a critical first step in the pipeline of vaccine production. This vaccine could be a powerful tool in our fight against virus-caused cancers. I hope you will agree to fund our ongoing research.

Best,

-Naomi Belgrano
Vice President and CFO of Phylogen

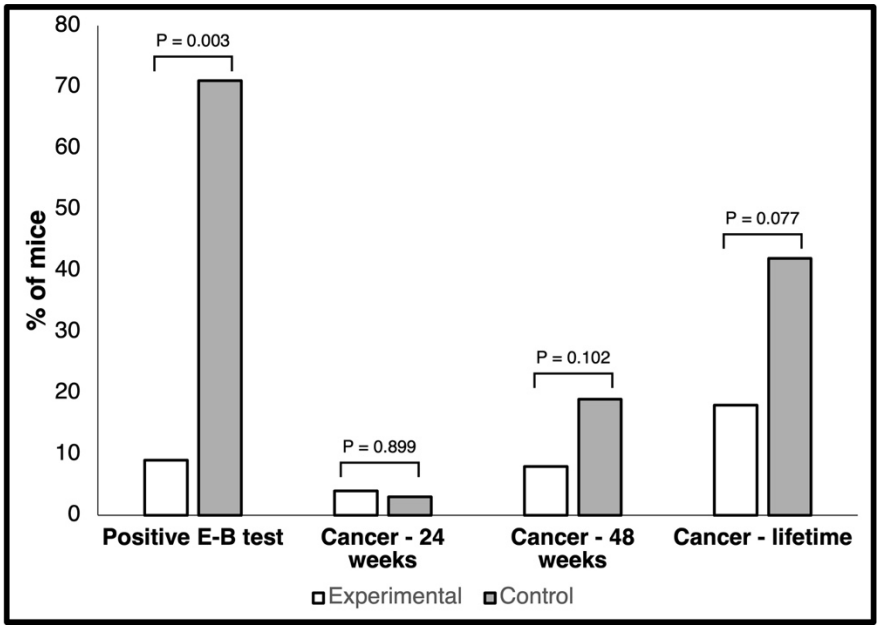


Figure 1. *Mice inoculated with either the PG-1aB vaccine (experimental) or saline (control) were exposed to Epstein-Barr virus at 10 days post-inoculation. A swab test for Epstein-Barr was performed 5 days later. Additionally, the number of mice developing cancers (of any type) was recorded at 24 weeks post-inoculation, 48 weeks post-inoculation, and for the remainder of their lifetimes.*

Andrew - can you put those p-values into the figure caption instead of over the bars? I'd rather the idiots over at the funding agency didn't see those right off the bat. And don't send over this version with my note on it. We don't want to look like dummies again. Remember what happened with Flutinimab (marketed in Canada as Nosecleanero).

-MB



The future of personalized cancer therapy is **here**, **now**. Genetechnic is excited to present its latest innovation – OncoCut™, a CRISPR-based cancer therapy that inactivates oncogenes and reduces cancer growth.

We are seeking **$1,800,000** to begin a series of lab animal trials. We will be using mice inoculated with cancer cells to test the efficacy of OncoCut™ in a live animal model. We anticipate that trials will take 18 months, at which time we hope to file for approval to begin Phase I testing in human subjects.

OncoCut is the first functional Oncogene Targeted Deactivation Therapy (OTDT™) to date. It harnesses the precision and power of CRISPR, along with modern advances in rapid genetic sequencing and Genetechnic's proprietary lipid nanosphere delivery system.

Here's how it works:

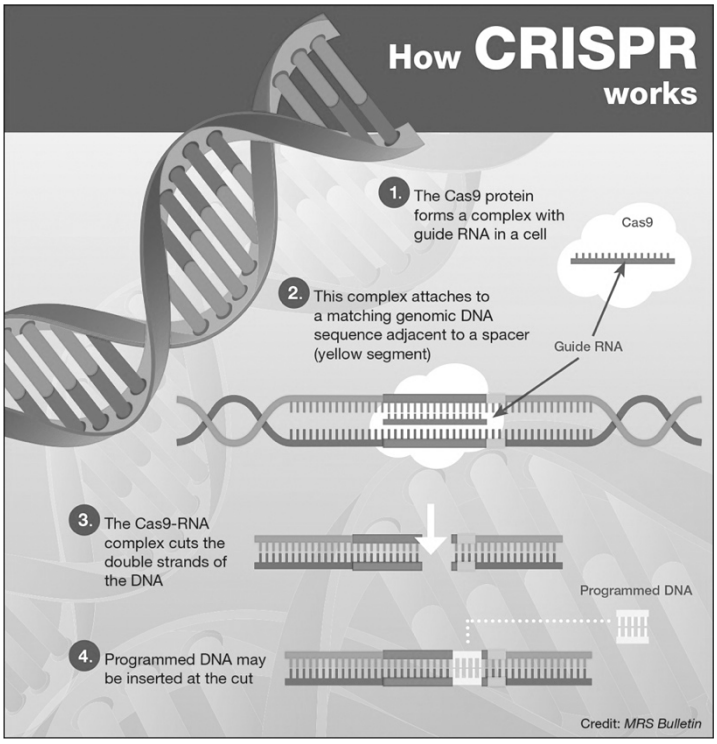
Step 1: A biopsy of the tumor is taken, and the cancer is sequenced.

Step 2: The site of the mutation is identified, and an 8 nucleotide-long guide RNA is produced which covers the region containing the mutation.

Step 3: Using our patented lipid nanosphere delivery system, Cas9 proteins with tailor-made guide RNAs are introduced into the tumor site along with copies of our proprietary OncoStop™ programmed donor DNA sequence.

Step 4: The guide RNA localizes the Cas9 complex to the target oncogene, which is then cut at the location of the mutation and the sequence from our donor DNA is added.

Step 5: Deactivation of the oncogene slows the replication rate of the cancer cells. This process may be repeated for other oncogenes identified from the original sample.



Lab trials with carcinoma cells in an aqueous medium showed a 99.998% success rate at editing the target region. And because the therapy targets the oncogene region that contains the cancer-causing mutation, there is almost no chance that these genes will be deactivated in healthy non-cancer cells.

If our trials are successful, OncoCut™ will be a valuable therapeutic for use with cancers of any kind and in any tissue. Delivery remains the primary hurdle – lipid nanospheres can be delivered by injection to most major tissue areas but are not yet able to move through the bloodstream or pass the blood-brain barrier.

Nonetheless, CRISPR-based cancer therapies open the door to personalized medicine, medicine that is tailor-made to fight **YOUR** cancer. If you have any questions about this proposal, please feel free to contact me.

Introducing OncoStop™ – a proprietary programmed donor DNA sequence:
ATTCATTCATT

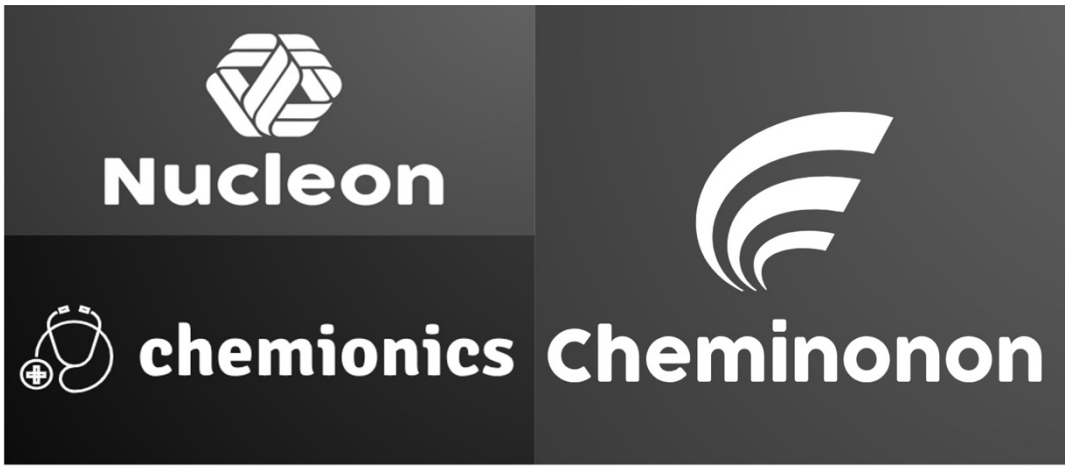
Margaret Thatcher
(no, not that one)

Genetechnic CEO and founder

Susan – quick note to say please check on the following info before our proposal goes out.

How long is the human genome? I mean, how many base pairs total?
What is the estimated probability of off-target localization? And how big a deal would that be? I mean I know it's a headache, but we should know the odds, right?

Also, don't forget to take these notes out before sending the proposal along. Remember what happened when Phylogen sent out that Flutinimab (marketed in Canada as Nosecleanero) info sheet with the notes left in?



Chemionics and Nucleon are excited to announce their merger into **Cheminonon**. This merger leverages the genetic engineering facilities of Nucleon and the skillsets of its modern and innovative staff alongside the more than 33-year-long success story of Chemionics's chemotherapy drug development program and the wisdom that only decades of delivering proven results can bring.

We are thrilled for the challenges that the future holds and are confident that our two teams will work as one to build a stronger, healthier tomorrow.

We are contacting you to request funding for a 2-year study into the efficacy of a novel cancer treatment. For this study, we are asking for $1,500,000 to cover research, consumables, and salary expenses.

Team Chemionics and **Team Nucleon** have each put together proposals based on their most current research. Currently, however, Cheminonon only has the resources to pursue one of these two paths. If you are willing to fund our ongoing research, please specify which of the two avenues you are choosing to fund. The Cheminonon family is in no way competitive, and each team is fully prepared to cast their project (and the years of hard, back-breaking work that went into it) into the dustbin to pursue whichever of the two you decide to support.

What are we offering?

Nucleon has developed a small chemical compound that binds to the eIF4E protein and reduces the rates of mRNA translation

Chemionics has developed a chemotherapy drug that binds to DNA polymerase and introduces multiple errors during DNA replication

Cheminonon is excited to pursue either of these two avenues. In the spirit of candid reflection, each team has critiqued the approach of the other and developed a brief technical report. This report is intended to be legible to a non-expert audience, since your understanding of the technique is critical to your decision-making regarding funding (and of course neither team would wish to sabotage the other).

Nucleon – *EG3f-Aie Small Particle translation inhibitor*

The binding of the eIF4E protein to the mRNA 5′ cap is a rate-limiting step in mRNA translation. As a result, eIF4E is an oncogene with significant up-regulation in many cancer lineages. Cells undergoing rapid division have greater quantities of protein to produce. By disrupting this binding, EG3F-Aie starves cancer cells of needed proteins, but might not severely impact normally dividing or non-dividing cells.

Pros – Mouse model research indicates that reductions in eIF4E are not associated with any developmental or physiological defects but are associated with a significant reduction in the likelihood of developing cancer. Side effects appear to be minimal, and there is reduced risk of hair loss, immunosuppression, and secondary cancers.

Cons – Inhibiting eIF4E function in all cells reduces translation rates across the board, with unclear consequences for long-term human health. All cells, including those not undergoing division will likely be affected. It is also unclear whether high translation rates are associated only with cancer cells, as well as whether all cancer types exhibit increased rates of protein production.

Chemionics – *Polymerase-specific binding factor TTa-1b(f)*

Reproducing cells must copy their genomes before dividing. Chemionics has developed a protein-based binding factor that targets DNA polymerase and disrupts replication. TTa-1b(f) binding reduces DNA polymerase accuracy, thus introducing tens of millions of mutations into the replicating genome. This leads to a breakdown in cellular function and eventual apoptosis (programmed cell death).

Pros – This approach is significantly more effective than Nucleon's molecule at specifically targeting dividing cell lines. Preliminary research indicates that tumors treated with Chemionics TTa-1b(f) shrink up to 210% more than those treated with Nucleon EG3f-Aie. The multiple-mutation technique ensures significant mortality for dividing cell lineages of all kinds, regardless of their specific suite of mutations or other characteristics.

Cons – At low dosages, mutations can be introduced to healthy cell tissues which may raise the risk of developing secondary cancers. This approach also targets all dividing cells, which includes many lineages of healthy cells (e.g., blood cells, epithelial cells, hair follicles).

We look forward to hearing from you and hope you are ready to move ahead with one of these innovative and promising cancer therapy approaches!

HEALTHLINE NYC

Press release – for immediate circulation:

For over 16 years Healthline NYC has served as New York City's strongest advocate for healthcare and healthy life choices. During that time, we have raised over $6 million in donations which have funded more than 20 different public service campaigns.

Our mission is to promote science-backed strategies for healthy living.

Building on the success of our 2008 advertisement campaign against sugary drinks – *"Just say no to H20 (that has sugar in it)"* – we have grown a team of advertising, research, and design experts that now puts together almost one campaign per year.

We are unique in that we combine our advertising, marketing, and design skills with unparalleled original research into messaging efficacy.

For 2025 we propose two public service campaigns intended to persuade New Yorkers to take cancer seriously:

1) ***Screen it!*** Research shows that sunscreen effectively protects against skin cancer, but less than 25% of New Yorkers report using sunscreen daily between March and October. We aim to bring that number WAY up!

2) ***Smoke is no joke!*** Smoking (and vaping, probably, we don't really know) are bad for your brain and for your body. Despite years of evidence linking tobacco use to lung cancer, esophageal cancer and mouth cancer, almost 45% of New Yorkers between the ages of 14 and 21 report using a tobacco product at least twice a week. That's up 50% from 2010!! We're going to clue kids into the cancer risks of smoking (and vaping? We're really not sure) and see if we can turn that trend upside-down.

We are asking for $1,000,000 total for these two ad campaigns, which will include bus and subway billboards, YouTube and broadcast television ad buys, radio spots, and COOL free posters handed out at public middle and high schools. ***We would accept $500,000 but would only be able to run one of these campaigns in that case.***

We have run preliminary tests on the efficacy of both ad campaigns. The results of these are listed below. Participants in the **Screen It!** Study were adults between the ages of 21 and 48 and participants in the **Smoke is no Joke!** Campaign were high school students between the ages of 16 and 18.

Participants (n = 36) were shown a series of PSAs from our proposed **Screen It!** Campaign. Before and after viewing the campaign, we asked participants to report the likelihood (on a scale of 1 to 10) that they would use sunscreen daily between the months of May and September.

	Mean	SD
Before Viewing	3.5	2.5
After Viewing	7.5	2

Participants (n = 11) were shown a series of PSAs from our proposed **Smoke is no Joke!** Campaign. Before and after viewing the campaign, we asked participants to report the likelihood (on a scale of 1 to 10) that they would use tobacco products at least twice a week in the next three months.

	Mean	SD
Before Viewing	6.64	3.3
After Viewing	3.21	2.6

We appreciate you taking the time to read our proposal! If you have any questions or comments, please reach out to our chief communications officer Derk Bellhoppe (dbelhoppppp@hlnyc.org)